

Stijn Velstra

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Keywords: algebraic and symplectic enumerative geometry, mathematical physics, cohomological field theories, moduli spaces of sections

Education

- Spring 2025 – 2028  **Ph.D. Mathematics**, Sorbonne Université Paris & Université Paul Sabatier Toulouse, CNRS.
Thesis title: *Reconstruction theorems in the enumerative geometry of spin curves* with Dr. A. Sauvaget and Dr. P. Georgieva.
- 2024  **M.Sc. Mathematics**, Leiden University, *Cum Laude*.
Thesis title: *The KP hierarchy and integrals on moduli spaces of curves and differentials*, with Dr. A. Sauvaget.
- 2020  **Double B.Sc. Mathematics and B.Sc. Physics**, Leiden University.
Thesis title: *The Berry connection and the Novikov problem: trajectories of Bloch electrons*, with Dr. F. Arici and Dr. V. Cheianov.

Projects

- Differentials   *Spin refinement of moduli spaces of residueless meromorphic differentials and the BKP hierarchy*, joint with D. Klompenhouwer, 2025
-  *Spin volumes of minimal strata via intersection theory*, joint with A. Bud and G. Poltopoulos, 2026

Activity

Seminars



- ◇ Enumerative geometry working group, Toulouse, 2026, *I will discuss **Hodge integrals** via localization and wrote some notes*
- ◇ Complex geometry conference, Albi, 2026, *I will discuss some **integrable hierarchies** in enumerative geometry*
- Seminar on **Hodge atoms and irrationality**, Paris, 2025
- **BKMP remodeling conjecture** working group, Toulouse, 2025
- Symplectic Dirty Hands on **Dehn-twists** and Floer homology, 2024
- ◇ Learning Seminar on **Log-Stacks**, Leiden and Utrecht University, 2024, *I introduced quasi-coherent sheaves and cohomology on stacks*
- ◇ **Arithmetic Deformation to the Normal Cone**, Leiden University, 2024, *I discussed Segre classes of cones and Gysin homomorphisms*
- Learning Seminar on **Stacks**, Leiden University, 2024
- **Variation of Hodge Structures**, Leiden University, 2024
- ◇ AG student seminar: **Toric Geometry**, Leiden University, 2024, *I discussed toric intersection theory and wrote accompanying lecture notes*
- ◇ **Hilbert Schemes of Points**, Leiden University, 2023-2024, *I discussed cobordism invariance of Hilbert schemes of points of a surface and updated a proof and some typos*
- **Mixed Hodge Structures**, Leiden University, 2023
- NCG seminar: **Non-Commutative Geometry of Foliations**, Leiden University, 2023
- AT student seminar: **Loop Spaces and Operads**, Utrecht University, Spring 2023
- ◇ FA student seminar: **Harmonic Analysis**, Leiden University, 2023, *I discussed Kirillov's Orbit Method and wrote accompanying lecture notes*
- **Differential Geometry of Moduli of Curves**, Leiden University, 2023
- NCG seminar: **Groupoid C*-Algebras**, Leiden University, 2022

Events



- Conference on **Fano** and **hyperKähler** manifolds, IMT, 2026
- ◇ CIRM conference on **Cycles on moduli spaces**, Marseille, 2025, *I gave a short talk on spin volumes and a Chiodo descendent*
- ◇ CAGE, Leiden University, 2025, *I discussed the intersection-theoretic computation of spin **Masur-Veech volumes** of minimal strata based on joint work with G. Politopoulos*
- Summer school on **Representations, Moduli and Duality**, EPFL, 2025
- **Spin structures** in Enumerative Geometry workshop, Leiden University, 2025
- New Trends in Moduli, **Integrability** and Deformations workshop, University of Padova, 2025
- ◇ Mathematics in Conversation seminars, University of Padova, 2025, *I was invited to give a talk on joint work with D. Klompenhouwer recovering the **BKP hierarchy** from spin-parity moduli spaces of differentials*
- Three day conference on **Isomonodromic foliations**, Université de Cergy-Pontoise, 2025
- Bourbaki seminar, 2025
- Conference on **Hodge Theory**, in honour of Chris Peters' 75th Birthday, Leiden University, 2024
- Workshop on **Enumerative Geometry and Arithmetic**, Lorentz Center, March 13, 2024
- **Poisson geometry** workshop Poisson aan de Waal, June 28, 2024
- Intercity seminar on **Hodge conjecture for categories**, March 8, April 5, 2024
- Belgian-Dutch Algebraic Geometry Seminar, Leiden University, October 27, April 26, 2023
- Three day conference on **Moduli Spaces of Vector Bundles and Curves**, Université de Cergy-Pontoise, 2023